

Introduction to Microeconomics

# Lecture 7 & 8—Government Interventions in the Market

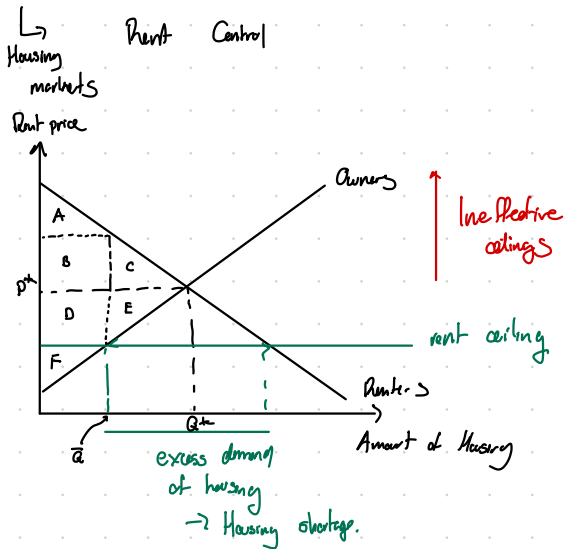
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Price Ceiling: max. price allowed legally to be charged



Surplus:  $A+B+C+D+E+F$

Without Ceiling Consumer Surplus  
 $A+B+C$

Producer Surplus  
 $D+E+F$

With rent Ceiling

Owners get at least: F

Renters ( $Q_c$ ): at least A

If there is a Black Market  
 $B+D$   
↕ If no Black Market  
goes to the consumer

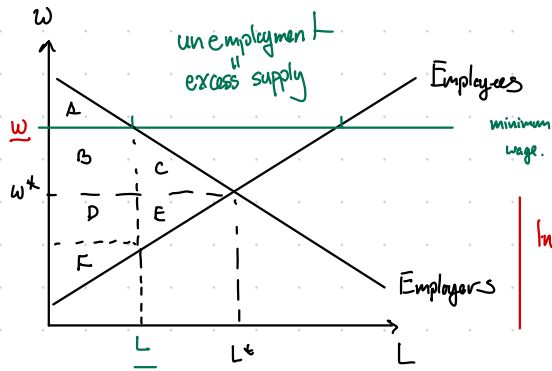
Search costs ←  $B+D$   
so consumer's may lose this amount.

DWL:  $C+E$

# Price floor:

L  
Labor Markets

minimum wage.



ineffective  
min  
wages

|    | No restriction  | Min wage                                                                            |
|----|-----------------|-------------------------------------------------------------------------------------|
| CS | $A+B+C$         | $A$ <span style="color:red">↑ Black Market</span>                                   |
| PS | $D+E+F$         | $F$ <span style="color:red">B+D</span> <span style="color:red">→ Search cost</span> |
| SS | $A+B+C + D+E+F$ | $A+B+D+F$                                                                           |

$$DWL = C+E$$

↑ Nobody appropriates this part of the surplus.

↓ creates an inefficiency.

# Taxes & Subsidies

## 1. Taxes (Per-Unit Tax)

Demand:  $q_D = Q_D(p^d)$   $p^d$  price consumers pay

Supply:  $q_S = Q_S(p^s)$   $p^s$  price producers receive.

In the absence of restriction:

↳ Market Equilibrium  $p^E = p^d = p^s$

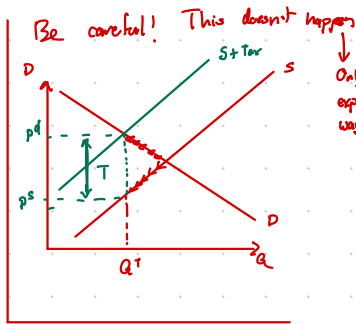
$$Q_D(p^E) = Q_S(p^E) = Q^E$$

With a per-unit tax: introduces a wedge between  $p^s$  and  $p^d$

$$\Rightarrow p^s \neq p^d ; \quad T = p^d - p^s$$

$$Q_D(p^d) = Q_S(p^s)$$

} This together  
identifies  
 $Q^T, p^d, p^s$



Assuming linear demand & supply:

$$Q_D = a - b P^D \quad Q_S = c + d P^S, \quad a, b, d > 0 \\ c \in \mathbb{R}$$

The tax imposed is  $T := P^D - P^S$

①  $P^S = P^D - T \Rightarrow$  "Auxiliary shift in the supply"

②  $P^D = T + P^S \Rightarrow$  "Auxiliary shift in the demand"

$$\rightarrow Q_S(P^D - T) = Q_D(P^D)$$

$$c + d(P^D - T) = a - b P^D$$

$$P^D = \frac{a - c + dT}{b + d}$$

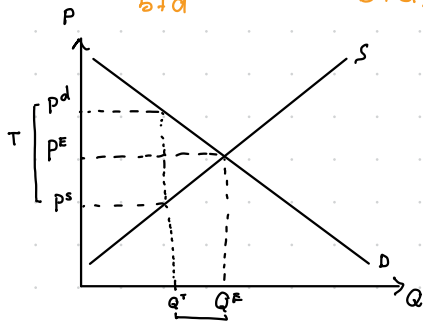
$$P^S = P^D - T = \frac{a - c + dT}{b + d} - T = \frac{a - c - bT}{b + d}$$

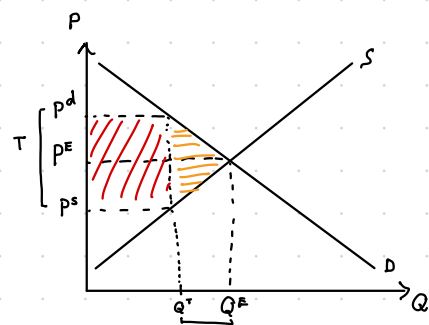
Plug the respective price in or the supply.

$$Q^T = a - b \left( \frac{a - c + dT}{b + d} \right) = c + d \left( \frac{a - c - bT}{b + d} \right) \\ = \frac{ad + bc - bdT}{b + d} \quad \text{eq. } Q \text{ after Tax.}$$

Note:  $T = 0$ , we recover the unrestricted eq. quantities:

$$P^E = \frac{a - c}{b + d}; \quad Q^E = \frac{ad + bc}{b + d}.$$





Government Revenue:  $T \cdot Q^T = T \times \left( \frac{ad + bc - bdT}{b+d} \right)$

DWL =  $\frac{(Q^E - Q^T) \times T}{2} = \frac{bdT^2}{2(b+d)}$

$DWL \propto T^2$   
 $\uparrow$   
 proportional

↳ If  $T > 0$ ,  $DWL > 0$  except when  $b$  or  $d$  are zero (Perfectly Inelastic cases)

Tax Incidence

$$p^E = \frac{a-c}{b+d}$$

| Consumers:                 | Incidence                                    |
|----------------------------|----------------------------------------------|
| $p^d = \frac{a-c+dT}{b+d}$ | $\frac{p^d - p^E}{T} = \frac{d}{b+d} = TI_D$ |
| $p^s = \frac{a-c-bT}{b+d}$ | $\frac{p^E - p^s}{T} = \frac{b}{b+d} = TI_S$ |
|                            | $TI_D + TI_S = 1$                            |

$$TI_D = \frac{d}{b+d} \cdot \frac{P/Q}{P/Q} = \frac{\epsilon_s^P}{|\epsilon_D^P| + \epsilon_s^P}$$

$$TI_S = \frac{b}{b+d} \cdot \frac{P/Q}{P/Q} = \frac{|\epsilon_D^P|}{|\epsilon_D^P| + \epsilon_s^P}$$

$TI_D > TI_S$   
 $\Leftrightarrow$  Consumers pay more when the supply is relatively more elastic

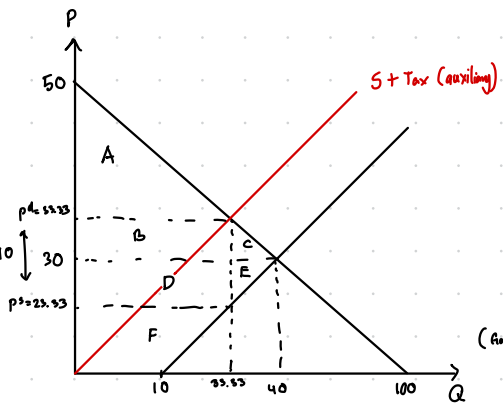
$$\frac{\epsilon_s^P}{|\epsilon_D^P| + \epsilon_s^P} > \frac{|\epsilon_D^P|}{|\epsilon_D^P| + \epsilon_s^P}$$

$$\Leftrightarrow \epsilon_s^P > |\epsilon_D^P| \Leftrightarrow \frac{\epsilon_s^P}{|\epsilon_D^P|} > 1$$

$TI_D = 0 \Leftrightarrow \epsilon_s^P = 0$   
 $\hookrightarrow$  Perfectly inelastic supply

$TI_S = 0 \Leftrightarrow |\epsilon_D^P| = 0$   
 Perfectly inelastic demand.

$Q_D = 100 - 2P^D$ ;  $Q_S = 10 + P^S$ ;  $T = 10$   
 $10 = P^D - P^S \Rightarrow P^S = P^D - 10$   
 $\hookrightarrow Q_S = 10 + P^D - 10 = P^D \Rightarrow Q_S = P^D$   
 $\Rightarrow 100 - 2P^D = P^D \Rightarrow P^D = \frac{100}{3} \approx 33.33$   $Q^T = 33.33$   
 $P^S \approx 23.33$   
 If  $T=0$ ;  $P^e = \frac{100-10}{2+1} = 30$ ,  $Q^e = 40$



|                                 | Area              | Amount                    |
|---------------------------------|-------------------|---------------------------|
| Tax paid by consumers           | B                 | $(P^D - P^E)Q^T = 110.99$ |
| Tax paid by producers           | D                 | $(P^E - P^S)Q^T = 222.34$ |
| Total Government Tax Collection | B + D             | $T \cdot Q^T = 333.33$    |
|                                 | Consumer Surplus  | Producer Surplus          |
| Before Tax                      | $A + B + C = 400$ | $D + E + F = 750$         |
| After Tax                       | $A = 277.81$      | $F = 505.47$              |

$(CS+PS)_{\text{before}} = A+B+C+D+E+F = 1150$   
 $(CS+PS)_{\text{after}} = A+F = 783.28$   
 $(\text{Government})R = 333.33$   
 $DWL = C+E = 33.35$

$(CS+PS)_{\text{before}} = (CS+PS)_{\text{after}} + R + DWL$   
 $(CS+PS)_{\text{before}} - (CS+PS)_{\text{after}} = R + DWL$   
 $\Rightarrow R + DWL$

Goes to government but  
 DWL nobody appropriates!



Subsidies A negative tax  $(-T)$

$$p^d - p^s = -T \Rightarrow \underline{p^d + S = p^s}$$

If we call

$$S = T$$

"Shifting the Supply"

"Shift" the demand  
 $p^d = p^s - S$

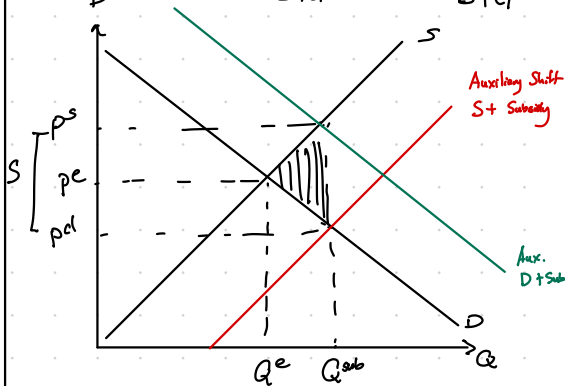
$$Q_s = c + d(p^d + S) \quad | \quad Q_d = a - b(p^s - S)$$

$$\hookrightarrow c + d(p^d + S) = a - b p^d$$

$$p^d = \frac{a - c - dS}{b + d}, \quad p^s = \frac{a - c + bS}{b + d}$$

$$Q^{\text{subsidy}} = \frac{ad + bc + dbS}{b + d}$$

$$S = 0, \quad p^E = \frac{a - c}{b + d}, \quad Q^E = \frac{ad + bc}{b + d}$$



$$\text{Gov. Expenditures} = S \times Q^{\text{sub}}$$

$$\text{DWL} = \frac{(Q^{\text{sub}} - Q^E)S}{2} = \frac{1}{2} \frac{db}{b + d} S^2$$

$$\text{Consumers Incidence: } \frac{d}{b + d} \Rightarrow \frac{\epsilon_s^D}{|\epsilon_s^D| + \epsilon_s^S}$$

$$\text{Producers Incidence: } \frac{b}{b + d} \Rightarrow \frac{|\epsilon_s^D|}{|\epsilon_s^D| + \epsilon_s^S}$$

$$Q_D = 100 - 2P^D; Q_S = 10 + P^S, S = 10$$

Without S:  $P^D = P^S = P^E = 30, Q^E = 40$

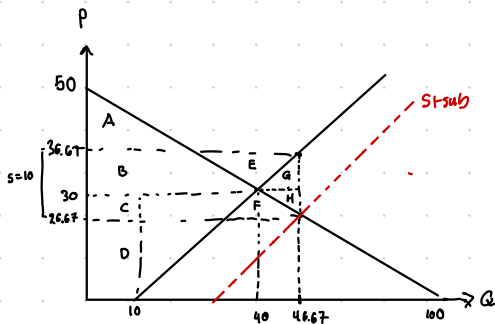
$$P^S - P^D = 10 \quad (P^D - P^S = -10)$$

$$P^S = 10 + P^D$$

$$\hookrightarrow 10 + 10 + P^D = 100 - 2P^D$$

$$P^D = 26.67, \quad P^S = 36.67$$

$$Q^{sub} = 100 - 2(26.67) = 46.67$$



|                               | Area                     | Amount                    |
|-------------------------------|--------------------------|---------------------------|
| Subsidy received by consumers | $C + F + H$              | 155.411                   |
| Subsidy received by producers | $B + E + G$              | 311.29                    |
| Total Government Subsidy      | $B + C + E + F + G + H$  | 466.7                     |
|                               | Consumer Surplus         | Producer Surplus          |
| Before Subsidy                | $A + B = 400$            | $C + D = 750$             |
| After Subsidy                 | $A + B + C + F = 544.46$ | $B + E + C + D = 1039.04$ |

$$(CS + PS)^{before} = 1150$$

$$(CS + PS)^{after} = 1583.5$$

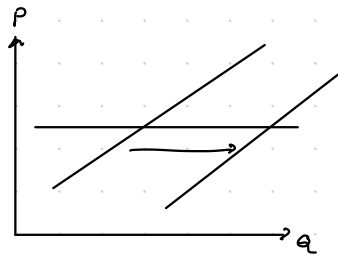
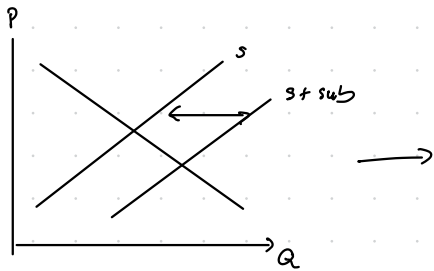
$$SS^{before} = 1150$$

$$SS^{after} = 1583.5 - E$$

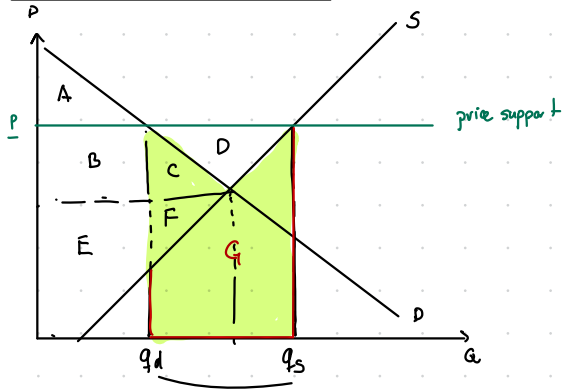
$$= 1583.5 - 466.7$$

$$DWL = 110.8 = G + H$$

Gov. Expenditure



# Agricultural Price Supports



$$q_d = Q_d(P)$$

$$q_s = Q_s(P)$$

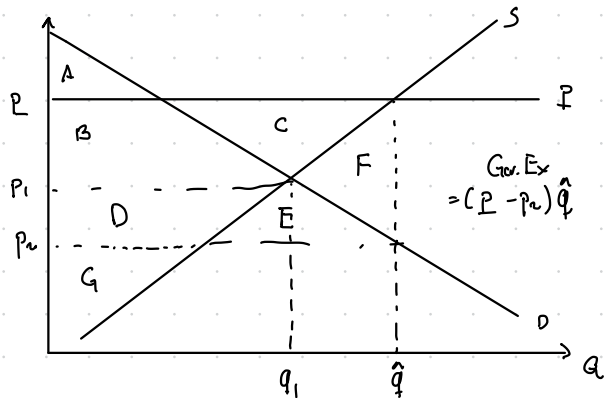
$$q_s - q_d = Q^{\text{excess}}$$

↳ Government will need to buy this in order to  $P \uparrow$

|     | No Price Support    | Price Support       | $\Delta$         |
|-----|---------------------|---------------------|------------------|
| CS  | $A + B + C$         | $A$                 | $-B - C$         |
| PS  | $E + F$             | $B + C + D + E + F$ | $B + C + D$      |
| Gov | $0$                 | $-C - D - F - G$    | $-C - D - F - G$ |
| SS  | $A + B + C + E + F$ | $A + B + E - G$     | $-C - F - G$     |

## Agricultural price supports: deficiency payments

- Gov put a price support of  $\underline{P} \rightarrow$  Producers: how much to produce at  $\underline{P}$ ? :  $\hat{q} = Q^s(\underline{P})$
- Gov. allows a price that clears the market :  $P_2 = (Q_0)^{-1}(\hat{q})$
- Gov. pay the difference :  $\underline{P} - P_2$  per unit sold.



|        | No Support      | Price Supp + Def. Payment | $\Delta$             |
|--------|-----------------|---------------------------|----------------------|
| CS     | $A + B$         | $A + B + D + E$           | $D + E$              |
| PS     | $D + G$         | $B + C + D + G$           | $B + C$              |
| Gov. X | 0               | $-B - C - D - E - F$      | $-B - C - D - E - F$ |
| SS     | $A + B + D + G$ | $A + B + D + G - F$       | $-F$                 |

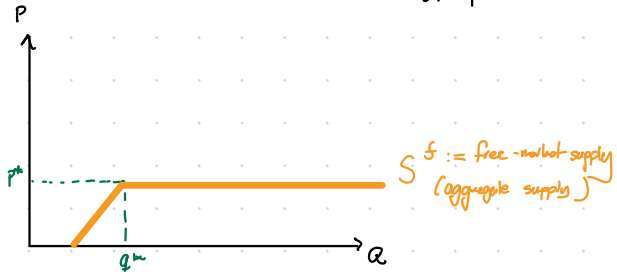
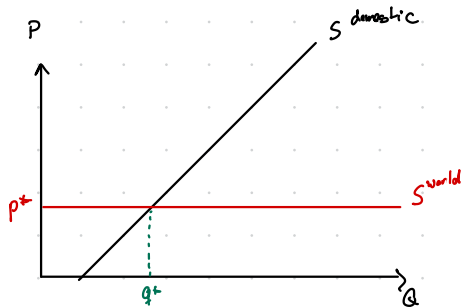
# Import Tariffs & Quotas

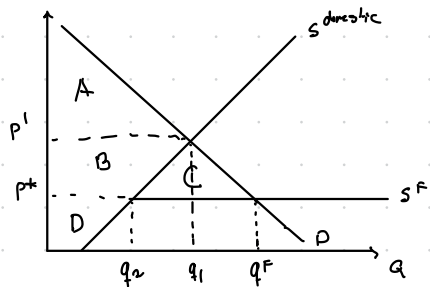
- A tariff is a tax that ONLY applies to imported goods.

- Assume :

- transportation costs are zero

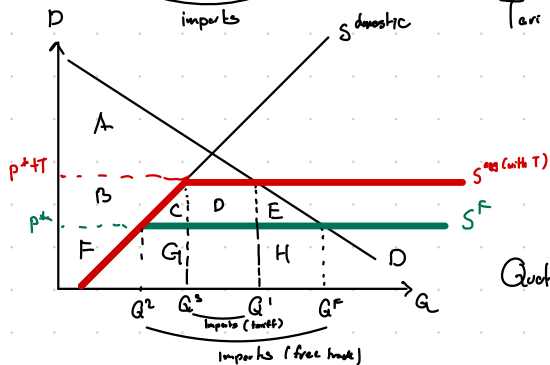
- The world price of the good under consideration is  $p^*$  } World Supply Curve perfectly elastic at  $p^*$





Total Ban on Imports

|    | FT        | Ban     | $\Delta$ |
|----|-----------|---------|----------|
| CS | $A+B+C$   | $A$     | $-B-C$   |
| PS | $D$       | $B+D$   | $B$      |
| SS | $A+B+C+D$ | $A+B+D$ | $-C$     |



Tariff T on Imports

|              | FT            | Tariff    | $\Delta$   |
|--------------|---------------|-----------|------------|
| CS           | $A+B+C+D+E$   | $A$       | $-B-C-D-E$ |
| PS           | $F$           | $B+F$     | $B$        |
| G (Revenues) | $0$           | $D$       | $D$        |
| SS           | $A+B+C+D+E+F$ | $A+B+D+F$ | $-C-E$     |

Quota on Imports?